

Jatin Swarnkar

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Experience

Software Engineer, Perceptiviti Data Solutions – Gurgaon, India July 2025 – Present

- Built an AI-powered **natural language-to-SQL platform** using **LangChain**, **Azure AI Foundry**, **Azure OpenAI**, and **PostgreSQL**, enabling analysts to query 25+ relational datasets in plain English and improving analyst efficiency by **60%**. Implemented context-aware schema retrieval, token-based conversational memory, self-healing SQL correction, and dynamic chart generation.
- Developed a configurable **case management system** with **role-based access control (RBAC)**, multi-stage audit workflows, bulk PostgreSQL processing, and concurrency-safe transitions using row-level locking.
- Developed and optimized backend APIs and database operations for the company's **analytics platform** using **Django REST Framework**, **PostgreSQL**, and modular backend design patterns, improving maintainability and system performance.
- Contributed to the company's **core fraud detection platform** built with **Django** and **PostgreSQL**, implementing backend features, optimizing database operations, and improving application performance.
- Enhanced Python-based **ETL pipelines**, enabling seamless extraction from **Oracle**, transformation, and multithreaded loading into **MySQL**.

Software Engineer Intern, Perceptiviti Data Solutions – Gurgaon, India Nov 2024 – June 2025

- Automated reporting using Python, reducing generation time from **2 days to under 2 hours** and eliminating repetitive manual tasks.
- Configured and deployed applications and databases across **AWS** and **Microsoft Azure**, working with **RDS**, **Azure VMs**, and **PostgreSQL Flexible Server** to improve deployment and data accessibility.
- Developed custom rules for **Sherlock's rule-based AI** using Python and MySQL, tailoring solutions to client requirements.

Key Skills

Skills: Python, C++, Django, Django REST Framework, LangChain, LangGraph, Azure AI Foundry, Azure OpenAI, MCP, REST APIs, SQL, PostgreSQL, MySQL, Oracle DB, FAISS, RAG, AWS, Microsoft Azure, ETL Pipelines, Multithreading, Git, Linux

Projects

Highlightly: AI-powered Video Highlights Generator Demo | GitHub

- Built an AI-powered web application that automatically generates highlight reels from YouTube videos using **OpenAI Whisper**, emotion analysis, scene detection, and audio peak analysis.
- Developed an end-to-end video processing pipeline with automated clipping, preview generation, and downloadable highlight outputs.
- **Tech:** Django, OpenAI Whisper, HuggingFace, FFmpeg, Tailwind CSS

AI Interview Copilot Demo | GitHub

- Built a multi-agent interview preparation platform using **LangGraph**, **Azure OpenAI**, and **FAISS**, performing resume analysis, job-description matching, skill-gap detection, and personalized interview preparation.
- Implemented **RAG-based retrieval**, semantic search, and agent orchestration for skill-gap analysis and adaptive interview coaching.
- **Tech:** LangGraph, Azure OpenAI, FAISS, Django, PostgreSQL, React

Education

Bennett University, B.Tech in Computer Science Engineering May 2025

- CGPA: 8.97